

Kirana Shop Management System - REST API Contracts

Authentication

All APIs require authentication. Include in headers:

```
Authorization: Bearer <jwt_token>  
Content-Type: application/json
```

The JWT token contains `retailer_id` - backend extracts this to filter all queries.

Screen 1: Purchase Management

1.1 Get All Suppliers

Endpoint: GET `/api/suppliers`

Headers:

```
Authorization: Bearer <token>
```

Query Parameters: - `page` (optional): Page number, default 1 - `limit` (optional): Items per page, default 20 - `search` (optional): Search by company name

Success Response (200):

```
{
  "success": true,
  "data": {
    "suppliers": [
      {
        "id": 1,
        "company_name": "ABC Distributors",
        "contact_person": "Ramesh Kumar",
        "phone": "9123456789",
        "email": "abc@suppliers.com",
        "address": "Warehouse 5, Industrial Area",
        "gst_number": "24AABCU9603R1ZM"
      }
    ],
    "pagination": {
      "total": 45,
      "page": 1,
      "limit": 20,
      "total_pages": 3
    }
  }
}
```

Error Response (401):

```
{
  "success": false,
  "error": {
    "code": "UNAUTHORIZED",
    "message": "Invalid or expired token"
  }
}
```

1.2 Add New Supplier

Endpoint: POST /api/suppliers

Headers:

```
Authorization: Bearer <token>
Content-Type: application/json
```

Request Payload:

```
{
  "company_name": "XYZ Wholesale",
  "contact_person": "Suresh Patel",
  "phone": "9876543210",
  "email": "xyz@suppliers.com",
  "address": "Plot 12, Market Yard",
  "gst_number": "27AABCU9603R1ZM"
}
```

Success Response (201):

```
{
  "success": true,
  "data": {
    "supplier": {
      "id": 46,
      "company_name": "XYZ Wholesale",
      "contact_person": "Suresh Patel",
      "phone": "9876543210",
      "email": "xyz@suppliers.com",
      "address": "Plot 12, Market Yard",
      "gst_number": "27AABCU9603R1ZM",
      "created_at": "2024-12-26T10:30:00Z"
    }
  },
  "message": "Supplier added successfully"
}
```

Error Response (400):

```
{
  "success": false,
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "Validation failed",
    "details": {
      "phone": "Phone number is required",
      "company_name": "Company name is required"
    }
  }
}
```

1.3 Search Items for Purchase

Endpoint: GET /api/items

Headers:

```
Authorization: Bearer <token>
```

Query Parameters: - `search` (optional): Search by name, SKU, or barcode - `category` (optional): Filter by category - `page` (optional): Page number - `limit` (optional): Items per page

Success Response (200):

```
{
  "success": true,
  "data": {
    "items": [
      {
        "id": 45,
        "name": "Tata Salt 1kg",
        "sku": "SALT-001",
        "barcode": "8901030123450",
        "category": "Groceries",
        "unit": "kg",
        "purchase_price": 35.00,
        "selling_price": 40.00,
        "current_stock": 50.5
      }
    ],
    "pagination": {
      "total": 150,
      "page": 1,
      "limit": 20,
      "total_pages": 8
    }
  }
}
```

1.4 Create Purchase Order

Endpoint: `POST /api/purchase-orders`

Headers:

```
Authorization: Bearer <token>
Content-Type: application/json
```

Request Payload:

```
{
  "supplier_id": 3,
  "order_date": "2024-12-26T10:00:00Z",
  "items": [
    {
      "item_id": 45,
      "quantity": 100,
      "unit_price": 35.00
    },
    {
      "item_id": 67,
      "quantity": 50,
      "unit_price": 120.00
    }
  ],
  "payment_status": "pending",
  "paid_amount": 0,
  "notes": "Delivery by truck"
}
```

Success Response (201):

```
{
  "success": true,
  "data": {
    "purchase_order": {
      "id": 123,
      "supplier_id": 3,
      "supplier_name": "ABC Distributors",
      "order_date": "2024-12-26T10:00:00Z",
      "status": "pending",
      "total_amount": 9500.00,
      "payment_status": "pending",
      "paid_amount": 0,
      "items": [
        {
          "id": 1,
          "item_id": 45,
          "item_name": "Tata Salt 1kg",
          "quantity": 100,
          "unit_price": 35.00,
          "amount": 3500.00
        },
        {
          "id": 2,
          "item_id": 67,
          "item_name": "Maggi Noodles",
          "quantity": 50,
          "unit_price": 120.00,
          "amount": 6000.00
        }
      ]
    }
  },
  "message": "Purchase order created successfully"
}
```

Error Response (400):

```
{
  "success": false,
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "Invalid request",
    "details": {
      "items": "At least one item is required",
      "supplier_id": "Invalid supplier ID"
    }
  }
}
```

Error Response (404):

```
{
  "success": false,
  "error": {
    "code": "NOT_FOUND",
    "message": "Supplier not found"
  }
}
```

1.5 Update Purchase Order Status

Endpoint: PATCH /api/purchase-orders/{id}/status

Headers:

```
Authorization: Bearer <token>
Content-Type: application/json
```

Request Payload:

```
{
  "status": "received"
}
```

Success Response (200):

```
{
  "success": true,
  "data": {
    "purchase_order": {
      "id": 123,
      "status": "received",
      "updated_at": "2024-12-26T15:30:00Z"
    }
  },
  "message": "Stock updated successfully"
}
```

Note: When status changes to "received", backend should: - Update `items.current_stock` for all items - Insert records in `inventory_transactions` - Update `items.last_purchase_date`

1.6 Add Item Master

Endpoint: POST /api/items

Headers:

```
Authorization: Bearer <token>
Content-Type: application/json
```

Request Payload:

```
{
  "name": "Tata Salt 1kg",
  "description": "Premium iodized salt",
  "sku": "SALT-001",
  "barcode": "8901030123450",
  "category": "Groceries",
  "unit": "kg",
  "purchase_price": 35.00,
  "selling_price": 40.00,
  "current_stock": 0,
  "min_stock_threshold": 10,
  "reorder_point": 15,
  "max_stock_level": 200
}
```

Success Response (201):

```
{
  "success": true,
  "data": {
    "item": {
      "id": 156,
      "name": "Tata Salt 1kg",
      "sku": "SALT-001",
      "barcode": "8901030123450",
      "category": "Groceries",
      "purchase_price": 35.00,
      "selling_price": 40.00,
      "current_stock": 0,
      "created_at": "2024-12-26T10:00:00Z"
    }
  },
  "message": "Item added successfully"
}
```

Error Response (409):

```
{
  "success": false,
  "error": {
    "code": "DUPLICATE",
    "message": "Item with this SKU or barcode already exists"
  }
}
```

Screen 2: Billing / Sales

2.1 Search Customer by Phone

Endpoint: GET /api/customers/search

Headers:

Authorization: Bearer <token>

Query Parameters: - `phone` : Phone number to search

Success Response (200):

```
{
  "success": true,
  "data": {
    "customer": {
      "id": 25,
      "name": "Priya Sharma",
      "phone_number": "9876543210",
      "email": "priya@gmail.com",
      "address": "Flat 201, Building A",
      "total_purchases": 15000.50
    }
  }
}
```

Error Response (404):

```
{
  "success": false,
  "error": {
    "code": "NOT_FOUND",
    "message": "Customer not found"
  }
}
```

2.2 Add New Customer

Endpoint: `POST /api/customers`

Headers:

Authorization: Bearer <token>
Content-Type: application/json

Request Payload:

```
{
  "name": "Rahul Verma",
  "phone_number": "9123456789",
  "email": "rahul@gmail.com",
  "address": "House 45, Sector 12"
}
```

Success Response (201):

```
{
  "success": true,
  "data": {
    "customer": {
      "id": 48,
      "name": "Rahul Verma",
      "phone_number": "9123456789",
      "email": "rahul@gmail.com",
      "address": "House 45, Sector 12",
      "total_purchases": 0,
      "created_at": "2024-12-26T11:00:00Z"
    }
  },
  "message": "Customer added successfully"
}
```

2.3 Search Items for Sale (with Stock Check)

Endpoint: GET /api/items/available

Headers:

```
Authorization: Bearer <token>
```

Query Parameters: - `search` : Search by name, SKU, or barcode - `category` (optional):
Filter by category

Success Response (200):

```
{
  "success": true,
  "data": {
    "items": [
      {
        "id": 45,
        "name": "Tata Salt 1kg",
        "sku": "SALT-001",
        "barcode": "8901030123450",
        "category": "Groceries",
        "unit": "kg",
        "selling_price": 40.00,
        "current_stock": 50.5,
        "is_available": true
      },
      {
        "id": 67,
        "name": "Maggi Noodles",
        "sku": "MAG-200",
        "selling_price": 12.00,
        "current_stock": 0,
        "is_available": false
      }
    ]
  }
}
```

2.4 Create Sale / Generate Bill

Endpoint: POST /api/sales

Headers:

```
Authorization: Bearer <token>
Content-Type: application/json
```

Request Payload:

```
{
  "customer_id": 25,
  "sale_date": "2024-12-26T14:30:00Z",
  "items": [
    {
      "item_id": 45,
      "quantity": 2
    },
    {
      "item_id": 67,
      "quantity": 5
    }
  ],
  "discount": 10.00,
  "payment_method": "upi",
  "payment_status": "paid",
  "paid_amount": 130.00,
  "notes": ""
}
```

Success Response (201):

```
{
  "success": true,
  "data": {
    "sale": {
      "id": 456,
      "bill_number": "BILL-456",
      "customer_id": 25,
      "customer_name": "Priya Sharma",
      "sale_date": "2024-12-26T14:30:00Z",
      "total_amount": 140.00,
      "discount": 10.00,
      "final_amount": 130.00,
      "payment_method": "upi",
      "payment_status": "paid",
      "paid_amount": 130.00,
      "items": [
        {
          "id": 1,
          "item_id": 45,
          "item_name": "Tata Salt 1kg",
          "quantity": 2,
          "unit_selling_price": 40.00,
          "unit_cost_price": 35.00,
          "amount": 80.00,
          "profit": 10.00
        },
        {
          "id": 2,
          "item_id": 67,
          "item_name": "Maggi Noodles",
          "quantity": 5,
          "unit_selling_price": 12.00,
          "unit_cost_price": 10.00,
          "amount": 60.00,
          "profit": 10.00
        }
      ],
      "total_profit": 20.00
    }
  },
  "message": "Bill created successfully"
}
```

Error Response (400 - Insufficient Stock):

```
{
  "success": false,
  "error": {
    "code": "INSUFFICIENT_STOCK",
    "message": "Insufficient stock for some items",
    "details": {
      "item_id": 67,
      "item_name": "Maggi Noodles",
      "requested": 5,
      "available": 3
    }
  }
}
```

Error Response (400 - Validation):

```
{
  "success": false,
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "Invalid request",
    "details": {
      "items": "At least one item is required"
    }
  }
}
```

Note: Backend should: - Validate stock availability for all items - Calculate profit using current purchase_price from items table - Update items.current_stock (decrease) - Insert into inventory_transactions - Update items.last_sale_date - Update customers.total_purchases if customer_id provided

2.5 Get Bill Details (for printing/viewing)

Endpoint: GET /api/sales/{id}

Headers:

```
Authorization: Bearer <token>
```

Success Response (200):

```
{
  "success": true,
  "data": {
    "sale": {
      "id": 456,
      "bill_number": "BILL-456",
      "customer": {
        "id": 25,
        "name": "Priya Sharma",
        "phone": "9876543210"
      },
      "sale_date": "2024-12-26T14:30:00Z",
      "items": [
        {
          "item_name": "Tata Salt 1kg",
          "quantity": 2,
          "unit_price": 40.00,
          "amount": 80.00
        },
        {
          "item_name": "Maggi Noodles",
          "quantity": 5,
          "unit_price": 12.00,
          "amount": 60.00
        }
      ],
      "total_amount": 140.00,
      "discount": 10.00,
      "final_amount": 130.00,
      "payment_method": "upi",
      "payment_status": "paid"
    },
    "retailer": {
      "business_name": "Sharma General Store",
      "address": "Shop 12, MG Road, Mumbai",
      "phone": "9876543210",
      "gst_number": "27AABCU9603R1ZM"
    }
  }
}
```

Screen 3: Stock Monitoring

3.1 Get All Items with Stock

Endpoint: GET /api/items/inventory

Headers:

```
Authorization: Bearer <token>
```

Query Parameters: - `category` (optional): Filter by category - `low_stock` (optional): true/false - show only low stock items - `page` (optional): Page number - `limit` (optional): Items

per page - `sort` (optional): sort field (name, current_stock, category) - `order` (optional): asc/desc

Success Response (200):

```
{
  "success": true,
  "data": {
    "items": [
      {
        "id": 45,
        "name": "Tata Salt 1kg",
        "sku": "SALT-001",
        "category": "Groceries",
        "unit": "kg",
        "purchase_price": 35.00,
        "selling_price": 40.00,
        "current_stock": 8.5,
        "min_stock_threshold": 10,
        "reorder_point": 15,
        "is_low_stock": true,
        "last_purchase_date": "2024-12-20T10:00:00Z",
        "last_sale_date": "2024-12-26T14:30:00Z"
      }
    ],
    "summary": {
      "total_items": 150,
      "low_stock_items": 12,
      "out_of_stock_items": 3,
      "total_stock_value": 125000.50
    },
    "pagination": {
      "total": 150,
      "page": 1,
      "limit": 50,
      "total_pages": 3
    }
  }
}
```

3.2 Get Low Stock Items

Endpoint: `GET /api/items/low-stock`

Headers:

```
Authorization: Bearer <token>
```

Success Response (200):

```
{
  "success": true,
  "data": {
    "items": [
      {
        "id": 45,
        "name": "Tata Salt 1kg",
        "current_stock": 8.5,
        "min_stock_threshold": 10,
        "reorder_point": 15,
        "shortage": 6.5,
        "last_purchase_date": "2024-12-20T10:00:00Z"
      }
    ],
    "total_count": 12
  }
}
```

3.3 Get Item Movement History

Endpoint: GET /api/items/{id}/transactions

Headers:

```
Authorization: Bearer <token>
```

Query Parameters: - `start_date` (optional): Filter from date - `end_date` (optional): Filter to date - `transaction_type` (optional): purchase/sale/adjustment/return - `page` (optional): Page number - `limit` (optional): Items per page

Success Response (200):

```

{
  "success": true,
  "data": {
    "item": {
      "id": 45,
      "name": "Tata Salt 1kg",
      "current_stock": 48.5
    },
    "transactions": [
      {
        "id": 1,
        "transaction_type": "purchase",
        "quantity": 100,
        "previous_stock": 0,
        "new_stock": 100,
        "reference_type": "purchase_order",
        "reference_id": 123,
        "notes": "Stock received from ABC Distributors",
        "created_at": "2024-12-20T10:00:00Z"
      },
      {
        "id": 2,
        "transaction_type": "sale",
        "quantity": -2,
        "previous_stock": 100,
        "new_stock": 98,
        "reference_type": "sale",
        "reference_id": 456,
        "created_at": "2024-12-26T14:30:00Z"
      },
      {
        "id": 3,
        "transaction_type": "adjustment",
        "quantity": -49.5,
        "previous_stock": 98,
        "new_stock": 48.5,
        "reference_type": "manual_adjustment",
        "notes": "Physical stock count correction",
        "created_by": "admin",
        "created_at": "2024-12-26T18:00:00Z"
      }
    ],
    "pagination": {
      "total": 25,
      "page": 1,
      "limit": 20,
      "total_pages": 2
    }
  }
}

```

3.4 Update Item Details

Endpoint: PATCH /api/items/{id}

Headers:

```
Authorization: Bearer <token>
Content-Type: application/json
```

Request Payload (all fields optional):

```
{
  "name": "Tata Salt 1kg Premium",
  "purchase_price": 36.00,
  "selling_price": 42.00,
  "min_stock_threshold": 15,
  "reorder_point": 20
}
```

Success Response (200):

```
{
  "success": true,
  "data": {
    "item": {
      "id": 45,
      "name": "Tata Salt 1kg Premium",
      "purchase_price": 36.00,
      "selling_price": 42.00,
      "min_stock_threshold": 15,
      "reorder_point": 20,
      "updated_at": "2024-12-26T16:00:00Z"
    }
  },
  "message": "Item updated successfully"
}
```

3.5 Manual Stock Adjustment

Endpoint: POST /api/items/{id}/adjust-stock

Headers:

```
Authorization: Bearer <token>
Content-Type: application/json
```

Request Payload:

```
{
  "new_stock": 48.5,
  "reason": "Physical stock count - found damaged packets"
}
```

Success Response (200):

```
{
  "success": true,
  "data": {
    "item": {
      "id": 45,
      "name": "Tata Salt 1kg",
      "previous_stock": 98.0,
      "new_stock": 48.5,
      "adjustment": -49.5
    },
    "transaction": {
      "id": 156,
      "transaction_type": "adjustment",
      "quantity": -49.5,
      "notes": "Physical stock count - found damaged packets",
      "created_at": "2024-12-26T18:00:00Z"
    }
  },
  "message": "Stock adjusted successfully"
}
```

Note: Backend should create `inventory_transactions` record with type='adjustment'

3.6 Get Categories

Endpoint: GET `/api/categories`

Headers:

```
Authorization: Bearer <token>
```

Success Response (200):

```
{
  "success": true,
  "data": {
    "categories": [
      {
        "name": "Groceries",
        "item_count": 85,
        "total_stock_value": 45000.00
      },
      {
        "name": "Snacks",
        "item_count": 42,
        "total_stock_value": 18000.00
      },
      {
        "name": "Beverages",
        "item_count": 23,
        "total_stock_value": 12000.00
      }
    ]
  }
}
```

Screen 4: Transaction History

4.1 Get Sales History

Endpoint: GET /api/sales

Headers:

```
Authorization: Bearer <token>
```

Query Parameters: - `start_date` (optional): From date - `end_date` (optional): To date - `customer_id` (optional): Filter by customer - `payment_status` (optional): paid/pending/partial - `payment_method` (optional): cash/card/upi/credit - `page` (optional): Page number - `limit` (optional): Items per page

Success Response (200):

```
{
  "success": true,
  "data": {
    "sales": [
      {
        "id": 456,
        "bill_number": "BILL-456",
        "customer_name": "Priya Sharma",
        "sale_date": "2024-12-26T14:30:00Z",
        "total_amount": 140.00,
        "discount": 10.00,
        "final_amount": 130.00,
        "payment_method": "upi",
        "payment_status": "paid",
        "items_count": 2
      }
    ],
    "summary": {
      "total_sales": 45000.00,
      "total_bills": 125,
      "total_profit": 8500.00,
      "cash_sales": 20000.00,
      "card_sales": 10000.00,
      "upi_sales": 15000.00,
      "pending_amount": 5000.00
    },
    "pagination": {
      "total": 125,
      "page": 1,
      "limit": 20,
      "total_pages": 7
    }
  }
}
```

4.2 Get Purchase History

Endpoint: GET /api/purchase-orders

Headers:

```
Authorization: Bearer <token>
```

Query Parameters: - `start_date` (optional): From date - `end_date` (optional): To date - `supplier_id` (optional): Filter by supplier - `status` (optional): pending/received/cancelled - `payment_status` (optional): pending/partial/paid - `page` (optional): Page number - `limit` (optional): Items per page

Success Response (200):

```
{
  "success": true,
  "data": {
    "purchase_orders": [
      {
        "id": 123,
        "supplier_name": "ABC Distributors",
        "order_date": "2024-12-20T10:00:00Z",
        "status": "received",
        "total_amount": 25000.00,
        "payment_status": "partial",
        "paid_amount": 10000.00,
        "pending_amount": 15000.00,
        "items_count": 5
      }
    ],
    "summary": {
      "total_purchases": 80000.00,
      "total_orders": 35,
      "pending_payment": 25000.00
    },
    "pagination": {
      "total": 35,
      "page": 1,
      "limit": 20,
      "total_pages": 2
    }
  }
}
```

4.3 Get Supplier Payment History

Endpoint: GET /api/suppliers/{id}/payments

Headers:

```
Authorization: Bearer <token>
```

Query Parameters: - start_date (optional): From date - end_date (optional): To date

Success Response (200):

```
{
  "success": true,
  "data": {
    "supplier": {
      "id": 3,
      "company_name": "ABC Distributors",
      "total_purchased": 150000.00,
      "total_paid": 120000.00,
      "pending_amount": 30000.00
    },
    "payments": [
      {
        "id": 1,
        "purchase_order_id": 123,
        "payment_date": "2024-12-22T16:00:00Z",
        "amount": 10000.00,
        "payment_method": "upi",
        "reference_number": "UTR987654321",
        "notes": "Partial payment"
      }
    ],
    "pending_orders": [
      {
        "id": 123,
        "order_date": "2024-12-20T10:00:00Z",
        "total_amount": 25000.00,
        "paid_amount": 10000.00,
        "pending_amount": 15000.00
      }
    ]
  }
}
```

4.4 Record Supplier Payment

Endpoint: POST /api/supplier-payments

Headers:

```
Authorization: Bearer <token>
Content-Type: application/json
```

Request Payload:

```
{
  "supplier_id": 3,
  "purchase_order_id": 123,
  "amount": 15000.00,
  "payment_method": "bank_transfer",
  "reference_number": "TXN123456789",
  "payment_date": "2024-12-26T10:00:00Z",
  "notes": "Final payment for PO-123"
}
```

Success Response (201):

```
{
  "success": true,
  "data": {
    "payment": {
      "id": 25,
      "supplier_id": 3,
      "supplier_name": "ABC Distributors",
      "purchase_order_id": 123,
      "amount": 15000.00,
      "payment_method": "bank_transfer",
      "reference_number": "TXN123456789",
      "payment_date": "2024-12-26T10:00:00Z",
      "created_at": "2024-12-26T10:05:00Z"
    },
    "purchase_order": {
      "id": 123,
      "total_amount": 25000.00,
      "paid_amount": 25000.00,
      "payment_status": "paid"
    }
  },
  "message": "Payment recorded successfully"
}
```

Note: Backend should update `purchase_orders.paid_amount` and `payment_status`

4.5 Get Pending Supplier Payments

Endpoint: `GET /api/supplier-payments/pending`

Headers:

```
Authorization: Bearer <token>
```

Success Response (200):

```
{
  "success": true,
  "data": {
    "pending_payments": [
      {
        "purchase_order_id": 123,
        "supplier_id": 3,
        "supplier_name": "ABC Distributors",
        "order_date": "2024-12-20T10:00:00Z",
        "total_amount": 25000.00,
        "paid_amount": 10000.00,
        "pending_amount": 15000.00,
        "days_pending": 6
      }
    ],
    "total_pending": 45000.00
  }
}
```

4.6 Get Daily/Monthly Reports

Endpoint: GET /api/reports/summary

Headers:

```
Authorization: Bearer <token>
```

Query Parameters: - type : daily/weekly/monthly - start_date : From date - end_date : To date

Success Response (200):

```
{
  "success": true,
  "data": {
    "report_type": "daily",
    "period": {
      "start_date": "2024-12-01",
      "end_date": "2024-12-26"
    },
    "summary": [
      {
        "date": "2024-12-26",
        "total_sales": 45000.00,
        "total_purchases": 25000.00,
        "total_profit": 8500.00,
        "total_bills": 125,
        "cash_sales": 20000.00,
        "card_sales": 10000.00,
        "upi_sales": 15000.00,
        "credit_sales": 5000.00
      },
      {
        "date": "2024-12-25",
        "total_sales": 38000.00,
        "total_purchases": 15000.00,
        "total_profit": 7200.00,
        "total_bills": 98,
        "cash_sales": 18000.00,
        "card_sales": 8000.00,
        "upi_sales": 12000.00,
        "credit_sales": 3000.00
      }
    ],
    "totals": {
      "total_sales": 450000.00,
      "total_purchases": 280000.00,
      "total_profit": 85000.00,
      "total_bills": 1250,
      "average_bill_value": 360.00
    }
  }
}
```

4.7 Get Top Selling Items

Endpoint: GET /api/reports/top-items

Headers:

```
Authorization: Bearer <token>
```

Query Parameters: - start_date : From date - end_date : To date - limit (optional): Number of items, default 10

Success Response (200):

```
{
  "success": true,
  "data": {
    "top_items": [
      {
        "item_id": 45,
        "item_name": "Tata Salt 1kg",
        "category": "Groceries",
        "total_quantity_sold": 250,
        "total_sales": 10000.00,
        "total_profit": 1250.00
      },
      {
        "item_id": 67,
        "item_name": "Maggi Noodles",
        "category": "Snacks",
        "total_quantity_sold": 500,
        "total_sales": 6000.00,
        "total_profit": 1000.00
      }
    ]
  }
}
```

Common Endpoints

Get Dashboard Summary

Endpoint: GET /api/dashboard

Headers:

```
Authorization: Bearer <token>
```

Success Response (200):

```
{
  "success": true,
  "data": {
    "today": {
      "sales": 45000.00,
      "bills": 125,
      "profit": 8500.00,
      "cash_sales": 20000.00,
      "digital_sales": 25000.00
    },
    "this_month": {
      "sales": 450000.00,
      "purchases": 280000.00,
      "profit": 85000.00,
      "bills": 1250
    },
    "inventory": {
      "total_items": 150,
      "low_stock_items": 12,
      "out_of_stock_items": 3,
      "total_stock_value": 125000.50
    },
    "payments": {
      "pending_supplier_payments": 45000.00,
      "pending_customer_payments": 15000.00
    }
  }
}
```

Error Response Format

All error responses follow this format:

400 - Bad Request:

```
{
  "success": false,
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "Validation failed",
    "details": {
      "field_name": "Error message"
    }
  }
}
```

401 - Unauthorized:

```
{
  "success": false,
  "error": {
    "code": "UNAUTHORIZED",
    "message": "Invalid or expired token"
  }
}
```

403 - Forbidden:

```
{
  "success": false,
  "error": {
    "code": "FORBIDDEN",
    "message": "Access denied to this resource"
  }
}
```

404 - Not Found:

```
{
  "success": false,
  "error": {
    "code": "NOT_FOUND",
    "message": "Resource not found"
  }
}
```

409 - Conflict:

```
{
  "success": false,
  "error": {
    "code": "DUPLICATE",
    "message": "Resource already exists"
  }
}
```

500 - Internal Server Error:

```
{
  "success": false,
  "error": {
    "code": "INTERNAL_ERROR",
    "message": "Something went wrong. Please try again later."
  }
}
```

Common Error Codes

Code	Description
UNAUTHORIZED	Invalid/expired token
FORBIDDEN	Retailer accessing another retailer's data
VALIDATION_ERROR	Invalid request payload
NOT_FOUND	Resource doesn't exist
DUPLICATE	Unique constraint violation
INSUFFICIENT_STOCK	Not enough stock for sale
INTERNAL_ERROR	Server error

Pagination Format

All list endpoints support pagination:

Query Parameters: - `page` : Page number (default: 1) - `limit` : Items per page (default: 20, max: 100)

Response includes:

```
{
  "pagination": {
    "total": 150,
    "page": 1,
    "limit": 20,
    "total_pages": 8
  }
}
```

Date Format

All dates use ISO 8601 format: - DateTime: `2024-12-26T14:30:00Z` (UTC) - Date only: `2024-12-26`

Notes for Frontend Team

- 1. **Authentication:** Store JWT token securely, include in all API calls
- 2. **Stock Validation:** Before creating sale, check `is_available` and `current_stock`
- 3. **Error Handling:** Always check `success` field, handle all error codes
- 4. **Pagination:** Implement infinite scroll or page navigation

5. **Real-time Updates:** Consider WebSocket for live stock updates (future enhancement)
6. **Offline Support:** Cache categories, items for offline billing (future enhancement)
7. **Print Support:** Use bill API response to generate print-ready invoice
8. **Number Formatting:** Format currency to 2 decimal places (₹10,000.50)
9. **Date Localization:** Convert UTC to local timezone for display
10. **Loading States:** Show loaders during API calls
11. **Optimistic UI:** Update UI immediately, rollback on error
12. **Barcode Scanner:** Integrate with search items API using barcode parameter